

CDS 101 Disney Project

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```
### Shows Each Column with examples
```

```
glimpse(disney)
```

```
## Rows: 1,450
## Columns: 12
## $ show_id      <chr> "s1", "s2", "s3", "s4", "s5", "s6", "s7", "s8", "s9", "s1~
## $ type         <chr> "Movie", "Movie", "Movie", "Movie", "TV Show", "Movie", "~
## $ title        <chr> "Duck the Halls: A Mickey Mouse Christmas Special", "Erne~
## $ director     <chr> "Alonso Ramirez Ramos, Dave Wasson", "John Cherry", "Kare~
## $ cast         <chr> "Chris Diamantopoulos, Tony Anselmo, Tress MacNeille, Bil~
## $ country      <chr> "", "", "United States", "", "", "United States", "", "Un~
## $ date_added   <chr> "November 26, 2021", "November 26, 2021", "November 26, 2~
## $ release_year <int> 2016, 1988, 2011, 2021, 2021, 2021, 2021, 2015, 2019, 200~
## $ rating       <chr> "TV-G", "PG", "TV-G", "TV-PG", "", "PG-13", "TV-14", "TV~
## $ duration     <chr> "23 min", "91 min", "23 min", "41 min", "1 Season", "94 m~
## $ listed_in    <chr> "Animation, Family", "Comedy", "Animation, Comedy, Family~
## $ description  <chr> "Join Mickey and the gang as they duck the halls!", "Sant~
```

```
### Shows each Variable
```

```
summary(disney)
```

```
##      show_id          type          title          director
## Length:1450      Length:1450      Length:1450      Length:1450
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##      cast          country          date_added          release_year
## Length:1450      Length:1450      Length:1450      Min.   :1928
## Class :character Class :character Class :character      1st Qu.:1999
## Mode  :character Mode  :character Mode  :character      Median :2011
##                                     Mean   :2003
##                                     3rd Qu.:2018
```

```
##
##      rating      duration      listed_in      Max.      :2021
## Length:1450    Length:1450    Length:1450    Length:1450
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
```

2.)

```
### Converts the release year variable from character to a numeric value
disney <- disney %>%
  mutate(release_year = as.numeric(release_year))
```

```
### Converts the duration variable from character to a numeric value
disney <- disney %>%
  mutate(duration_num = parse_number(duration))
```

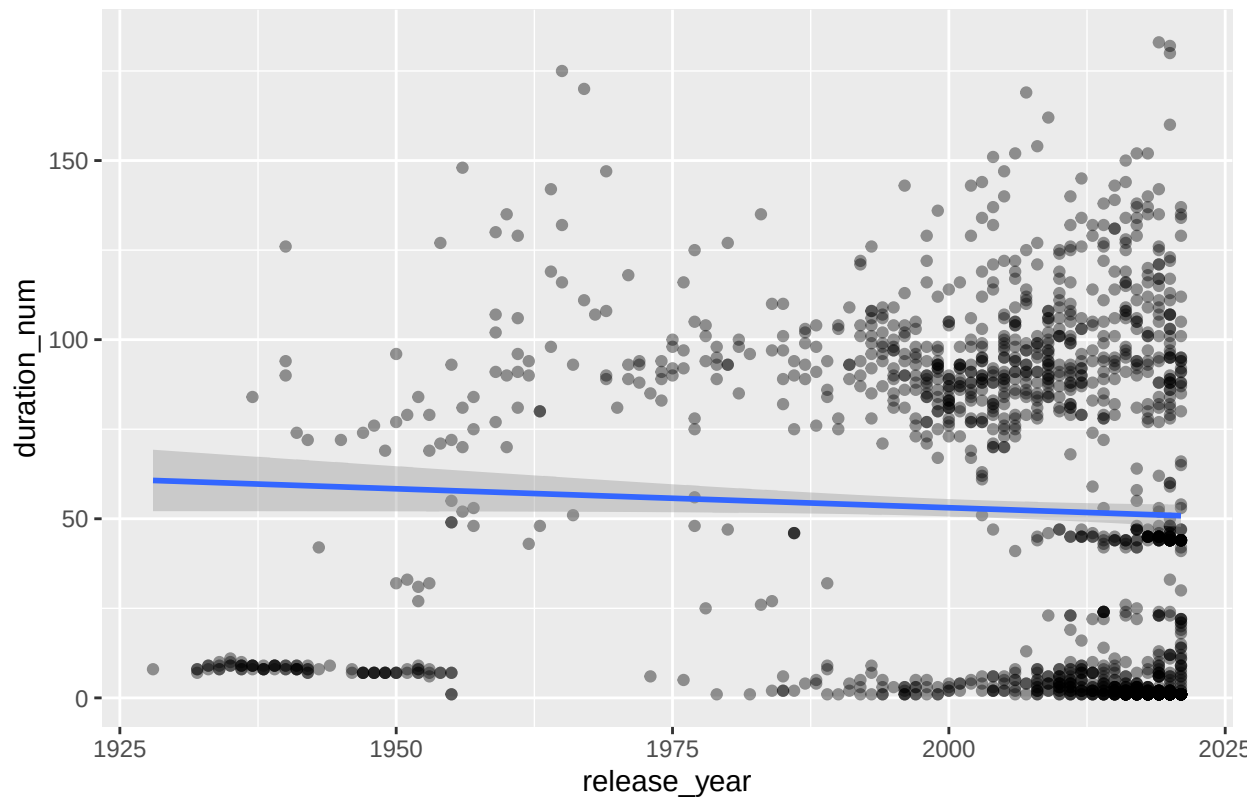
```
### Transforms the ratings and listed_in variables into factors which makes them categorical
disney <- disney %>%
  mutate(
    rating = as.factor(rating),
    listed_in = as.factor(listed_in)
  )
```

3.) Exploring Relationships:

```
#Creates a scatter plot looking at the relationship between the release year
#and the duration of the movie to see if there is any correlation between the two.
ggplot(disney, aes(x = release_year, y = duration_num)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm") +
  labs(title = "Duration vs Year of Release")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Duration vs Year of Release



ii.)

*# Adds a new column named period which splits release years into different periods
of time being 25 years apart from each other.*

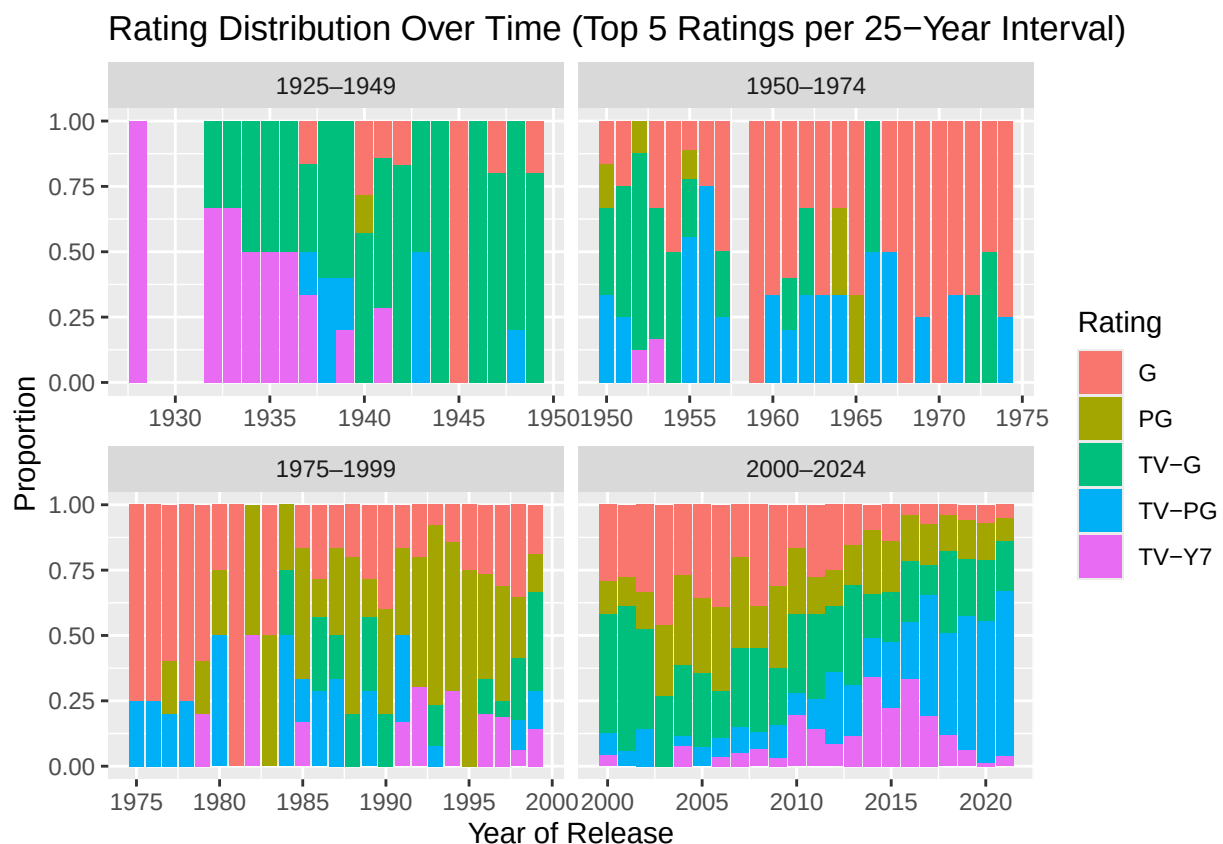
```
disney <- disney %>%
  mutate(period = cut(
    release_year,
    breaks = c(1900, 1925, 1950, 1975, 2000, 2025),
    labels = c("1900-1924", "1925-1949", "1950-1974", "1975-1999", "2000-2024"),
    right = FALSE
  ))
```

*# Creates a new data set looking at the amount of movies spread based on
two different factors.
The factors being time period and rating of the movie.
For example, This just gives the total amount of movies for rated G during
the years of 2000-2024*

```
top_ratings <- disney %>%
  count(period, rating, sort = TRUE) %>%
  group_by(period) %>%
  slice_max(n, n = 5) %>%
  ungroup()
```

```
#New dataset making sure all the values match from top_ratings using the semi_join
#function
ratings_top <- disney %>%
  semi_join(top_ratings, by = c("period", "rating"))
```

```
#Creates a bar plot looking at the distribution of age rating throughout the years.
ratings_top %>%
  count(period, release_year, rating) %>%
  ggplot(aes(x = release_year, y = n, fill = rating)) +
  geom_col(position = "fill") +
  facet_wrap(~ period, scales = "free_x") +
  labs(
    title = "Rating Distribution Over Time (Top 5 Ratings per 25-Year Interval)",
    x = "Year of Release",
    y = "Proportion",
    fill = "Rating"
  )
)
```



iii.)

```
# Creates a new data set with new rows based on the listed_in Variable.
# This new data set creates new rows based on the different genres of each movie.
```

```
# For example the first movie in the data set
# "Duck the Halls: A Mickey Mouse Christmas Special"
# is split into two different two rows with one having animation and the other being family.
# Not all rows will be split, but all rows will keep the other same data
# such as rating and release year.
```

```
disney_genre <- disney %>%
  separate_rows(listed_in, sep = ", ") %>%
  mutate(listed_in = as.factor(listed_in))
```

```
# This does the same thing as earlier, adding a new column named period.
# This add a column based on the release date of the movie,
#being 25 years apart from each other.
```

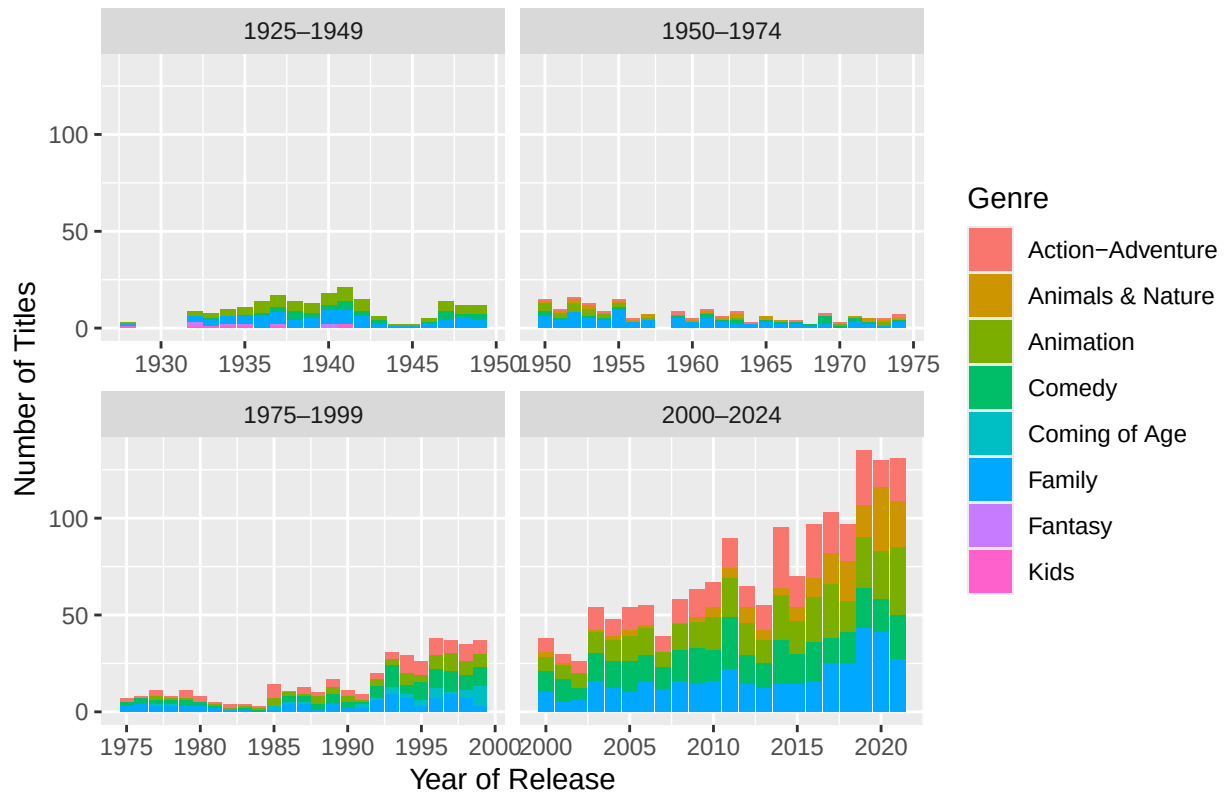
```
disney_genre <- disney_genre %>%
  mutate(period = cut(
    release_year,
    breaks = c(1900, 1925, 1950, 1975, 2000, 2025),
    labels = c("1900-1924", "1925-1949", "1950-1974", "1975-1999", "2000-2024"),
    right = FALSE
  ))
```

```
top_genres <- disney_genre %>%
  count(period, listed_in, sort = TRUE) %>%
  group_by(period) %>%
  slice_max(n, n = 5) %>%
  ungroup()
```

```
disney_top <- disney_genre %>%
  semi_join(top_genres, by = c("period", "listed_in"))
```

```
disney_top %>%
  count(period, release_year, listed_in) %>%
  ggplot(aes(x = release_year, y = n, fill = listed_in)) +
  geom_col() +
  facet_wrap(~ period, scales = "free_x") +
  labs(
    title = "Top 5 Genres per 25-Year Interval",
    x = "Year of Release",
    y = "Number of Titles",
    fill = "Genre"
  )
```

Top 5 Genres per 25-Year Interval



4.)

```
model_duration <- lm(duration_num ~ release_year, data = disney)
tidy(model_duration)
```

term	estimate	std.error	statistic	p.value
(Intercept)	264.6730503	111.9810820	2.363551	0.0182322
release_year	-0.1057961	0.0559008	-1.892569	0.0586146

```
glance(model_duration)
```

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual	nobs
0.0024675	0.0017786	46.516343	3.581815	0.05861461		-	15254.351	15270.193	133139	1448	1450
						7624.175					

```
model_rating <- lm(as.numeric(rating) ~ release_year, data = disney)
tidy(model_rating)
```

term	estimate	std.error	statistic	p.value
(Intercept)	-13.5763019	5.508818	-2.464467	0.0138371
release_year	0.0093943	0.002750	3.416123	0.0006528

5.)

```
model_multi <- lm(duration_num ~ release_year + rating + listed_in, data = disney)
tidy(model_multi)
```

term	estimate	std.error	statistic	p.value
(Intercept)	256.7160698	82.0478244	3.1288589	0.0018006
release_year	-	0.0376554	-	0.0005837
	0.1298756		3.4490523	
ratingG	44.9567097	19.5768813	2.2964184	0.0218374
ratingPG	58.8122928	19.5623930	3.0063956	0.0027030
ratingPG-13	80.1718936	19.9736079	4.0138914	0.0000637
ratingTV-14	9.8983607	19.5955230	0.5051338	0.6135651
ratingTV-G	6.5935709	19.5786190	0.3367741	0.7363508
ratingTV-PG	7.9753963	19.4143681	0.4107987	0.6812994
ratingTV-Y	-	20.1502627	-	0.5884580
	10.9059150		0.5412294	
ratingTV-Y7	-	19.6940199	-	0.5280403
	12.4308453		0.6311990	
ratingTV-Y7-FV	8.6294797	21.0412526	0.4101220	0.6817956
listed_inAction-Adventure, Animals & Nature, Comedy	52.5733934	33.3432004	1.5767351	0.1151412
listed_inAction-Adventure, Animals & Nature, Documentary	57.4323556	25.2020986	2.2788719	0.0228638
listed_inAction-Adventure, Animals & Nature, Docuseries	0.6930826	24.8341052	0.0279085	0.9777401
listed_inAction-Adventure, Animals & Nature, Drama	51.9560387	25.8496268	2.0099338	0.0446796
listed_inAction-Adventure, Animals & Nature, Family	46.6872962	24.6522841	1.8938325	0.0585068
listed_inAction-Adventure, Animation	49.7159207	26.3083986	1.8897357	0.0590535
listed_inAction-Adventure, Animation, Anime	14.1274203	26.3866838	0.5353996	0.5924807
listed_inAction-Adventure, Animation, Anthology	11.2891264	25.8005415	0.4375539	0.6617948
listed_inAction-Adventure, Animation, Buddy	31.1302192	25.0306155	1.2436857	0.2138776
listed_inAction-Adventure, Animation, Comedy	26.8683769	23.7187774	1.1327893	0.2575472
listed_inAction-Adventure, Animation, Coming of Age	45.1274428	24.7082250	1.8264138	0.0680562
listed_inAction-Adventure, Animation, Family	40.6280493	23.8968926	1.7001394	0.0893846
listed_inAction-Adventure, Animation, Fantasy	22.1605221	24.3581387	0.9097790	0.3631364
listed_inAction-Adventure, Animation, Kids	20.9852799	23.8950508	0.8782270	0.3800104
listed_inAction-Adventure, Animation, Musical	41.8897994	28.7920758	1.4549072	0.1459774

term	estimate	std.error	statistic	p.value
listed_inAction-Adventure, Animation, Science Fiction	16.7941762	25.8676240	0.6492354	0.5163204
listed_inAction-Adventure, Animation, Superhero	30.9033826	29.1331713	1.0607627	0.2890283
listed_inAction-Adventure, Biographical, Drama	45.4435178	33.3445092	1.3628486	0.1732064
listed_inAction-Adventure, Buddy, Comedy	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAction-Adventure, Buddy, Science Fiction	-	33.4932403	-	0.9254214
	3.1359097		0.0936281	
listed_inAction-Adventure, Comedy	90.8311199	33.2096829	2.7350794	0.0063358
listed_inAction-Adventure, Comedy, Coming of Age	41.7651564	24.3007175	1.7186800	0.0859512
listed_inAction-Adventure, Comedy, Crime	37.1383312	28.9142594	1.2844296	0.1992594
listed_inAction-Adventure, Comedy, Family	50.5400022	24.0272140	2.1034483	0.0356510
listed_inAction-Adventure, Comedy, Fantasy	60.8708424	26.3360597	2.3113117	0.0209980
listed_inAction-Adventure, Comedy, Kids	52.6909004	25.7925074	2.0428762	0.0413003
listed_inAction-Adventure, Comedy, Science Fiction	56.8641265	25.2786607	2.2494913	0.0246764
listed_inAction-Adventure, Comedy, Spy/Espionage	41.8721491	33.3324501	1.2561978	0.2093084
listed_inAction-Adventure, Coming of Age	89.5713688	33.2105368	2.6970768	0.0071010
listed_inAction-Adventure, Coming of Age, Drama	52.1848422	26.4100797	1.9759441	0.0484082
listed_inAction-Adventure, Coming of Age, Family	56.3801610	24.8755202	2.2664917	0.0236129
listed_inAction-Adventure, Coming of Age, Fantasy	86.8389211	33.3345328	2.6050739	0.0093082
listed_inAction-Adventure, Crime, Family	88.3841473	33.3945181	2.6466664	0.0082439
listed_inAction-Adventure, Crime, Fantasy	42.8202826	33.3242971	1.2849568	0.1990752
listed_inAction-Adventure, Crime, Science Fiction	60.2009307	33.5824158	1.7926325	0.0733037
listed_inAction-Adventure, Disaster, Drama	43.9411796	33.5835041	1.3084156	0.1910030
listed_inAction-Adventure, Documentary	42.2597511	33.2080390	1.2725759	0.2034348
listed_inAction-Adventure, Documentary, Historical	91.5273035	33.3311558	2.7459985	0.0061301
listed_inAction-Adventure, Documentary, Sports	98.3974279	33.3311604	2.9521153	0.0032224
listed_inAction-Adventure, Documentary, Superhero	39.8779257	33.3316039	1.1963998	0.2317958
listed_inAction-Adventure, Docuseries, Family	-	27.2754827	-	0.9429158
	1.9535187		0.0716218	
listed_inAction-Adventure, Docuseries, Fantasy	-	33.4932403	-	0.9254214
	3.1359097		0.0936281	
listed_inAction-Adventure, Docuseries, Historical	-	28.9011055	-	0.9593694
	1.4726965		0.0509564	
listed_inAction-Adventure, Docuseries, Lifestyle	-	33.4936600	-	0.9636795
	1.5255364		0.0455470	
listed_inAction-Adventure, Docuseries, Reality	-	33.4932403	-	0.9254214
	3.1359097		0.0936281	

term	estimate	std.error	statistic	p.value
listed_inAction-Adventure, Drama, Family	42.8833317	33.3770522	1.2848148	0.1991248
listed_inAction-Adventure, Drama, Historical	50.4216773	33.5861873	1.5012623	0.1335719
listed_inAction-Adventure, Family	18.1682639	27.2846674	0.6658782	0.5056272
listed_inAction-Adventure, Family, Fantasy	62.8101740	24.3474911	2.5797391	0.0100151
listed_inAction-Adventure, Family, Game Show / Competition	- 3.1359097	33.4932403	- 0.0936281	0.9254214
listed_inAction-Adventure, Family, Historical	32.3711001	28.9604751	1.1177683	0.2639076
listed_inAction-Adventure, Family, Mystery	70.9613408	33.3432794	2.1282052	0.0335397
listed_inAction-Adventure, Family, Parody	30.1319002	33.3308101	0.9040254	0.3661779
listed_inAction-Adventure, Family, Science Fiction	46.1674212	24.4480011	1.8883925	0.0592336
listed_inAction-Adventure, Family, Survival	30.4302372	33.4780592	0.9089606	0.3635681
listed_inAction-Adventure, Family, Western	45.7015896	33.3469076	1.3704896	0.1708110
listed_inAction-Adventure, Fantasy	58.9991209	24.8769456	2.3716385	0.0178791
listed_inAction-Adventure, Fantasy, Musical	72.0410292	33.3264098	2.1616799	0.0308559
listed_inAction-Adventure, Fantasy, Science Fiction	67.2088929	25.3499147	2.6512473	0.0081336
listed_inAction-Adventure, Fantasy, Soap Opera / Melodrama	0.6181745	28.9021724	0.0213885	0.9829396
listed_inAction-Adventure, Fantasy, Superhero	46.9411796	27.5728554	1.7024417	0.0889523
listed_inAction-Adventure, Fantasy, Survival	48.7015896	33.3469076	1.4604530	0.1444484
listed_inAction-Adventure, Mystery, Thriller	73.4370869	28.9021070	2.5408904	0.0111922
listed_inAction-Adventure, Reality, Survival	- 0.4726965	33.3311558	- 0.0141818	0.9886875
listed_inAction-Adventure, Science Fiction	66.1721683	24.4817512	2.7029181	0.0069782
listed_inAction-Adventure, Science Fiction, Spy/Espionage	28.7727593	28.9976340	0.9922451	0.3212940
listed_inAction-Adventure, Science Fiction, Superhero	52.5917722	24.9681188	2.1063570	0.0353972
listed_inAction-Adventure, Science Fiction, Thriller	39.4953843	33.3692333	1.1835868	0.2368298
listed_inAction-Adventure, Superhero	- 3.1359097	33.4932403	- 0.0936281	0.9254214
listed_inAction-Adventure, Western	40.5733934	33.3432004	1.2168416	0.2239229
listed_inAnimals & Nature	- 0.2129454	33.3312744	- 0.0063888	0.9949037
listed_inAnimals & Nature, Animation, Family	- 33.8436365	33.3064125	- 1.0161297	0.3097888
listed_inAnimals & Nature, Animation, Kids	- 1.0651471	33.2404100	- 0.0320437	0.9744429
listed_inAnimals & Nature, Anthology	0.1688801	33.2096829	0.0050853	0.9959435
listed_inAnimals & Nature, Biographical, Documentary	54.1265277	27.3148807	1.9815766	0.0477728
listed_inAnimals & Nature, Biographical, Drama	75.1632521	28.8111872	2.6088218	0.0092075
listed_inAnimals & Nature, Comedy	46.6514025	33.3280404	1.3997643	0.1618631

term	estimate	std.error	statistic	p.value
listed_inAnimals & Nature, Comedy, Coming of Age	85.3311199	28.7609282	2.9669112	0.0030725
listed_inAnimals & Nature, Comedy, Family	44.6258521	25.1959726	1.7711502	0.0768098
listed_inAnimals & Nature, Coming of Age, Drama	74.3296019	28.8606342	2.5754667	0.0101389
listed_inAnimals & Nature, Coming of Age, Family	85.4414932	33.2110278	2.5726844	0.0102202
listed_inAnimals & Nature, Documentary	45.1014933	23.9935138	1.8797369	0.0604055
listed_inAnimals & Nature, Documentary, Family	42.5968435	23.8821880	1.7836240	0.0747577
listed_inAnimals & Nature, Documentary, Medical	39.8250857	33.4952062	1.1889787	0.2347021
listed_inAnimals & Nature, Documentary, Survival	55.7342147	33.4933379	1.6640388	0.0963867
listed_inAnimals & Nature, Docuseries	-	24.7930762	-	0.9390485
	1.8962520		0.0764831	
listed_inAnimals & Nature, Docuseries, Family	-	23.8969174	-	0.9821692
	0.5342033		0.0223545	
listed_inAnimals & Nature, Docuseries, Kids	-	27.1149243	-	0.9909587
	0.3073303		0.0113344	
listed_inAnimals & Nature, Docuseries, Medical	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAnimals & Nature, Docuseries, Reality	-	25.9018327	-	0.9350281
	2.1120067		0.0815389	
listed_inAnimals & Nature, Docuseries, Survival	-	28.9015154	-	0.9432693
	2.0571366		0.0711775	
listed_inAnimals & Nature, Docuseries, Travel	-	33.3314292	-	0.9523501
	1.9921988		0.0597694	
listed_inAnimals & Nature, Drama, Family	63.8872202	25.4891510	2.5064476	0.0123371
listed_inAnimals & Nature, Family	3.0912163	33.3415289	0.0927137	0.9261477
listed_inAnimals & Nature, Family, Reality	41.0078012	33.3314292	1.2303043	0.2188436
listed_inAnimals & Nature, Family, Sports	80.2932763	33.3793247	2.4054794	0.0163139
listed_inAnimals & Nature, Kids, Music	-	33.2082952	-	0.9884582
	0.4804977		0.0144692	
listed_inAnimals & Nature, Kids, Variety	0.7143156	28.7596772	0.0248374	0.9801891
listed_inAnimals & Nature, Reality	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAnimation	3.1509257	29.0450521	0.1084841	0.9136313
listed_inAnimation, Anthology, Comedy	30.3031581	27.1296518	1.1169756	0.2642463
listed_inAnimation, Anthology, Family	3.2381200	26.3254045	0.1230036	0.9021265
listed_inAnimation, Anthology, Fantasy	28.8186033	33.3145815	0.8650447	0.3872010
listed_inAnimation, Anthology, Kids	19.7592370	33.6287067	0.5875705	0.5569400
listed_inAnimation, Anthology, Science Fiction	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAnimation, Buddy, Comedy	17.9752526	25.4417901	0.7065247	0.4800101
listed_inAnimation, Buddy, Coming of Age	33.5717272	28.8614987	1.1632011	0.2449977

term	estimate	std.error	statistic	p.value
listed_inAnimation, Buddy, Family	-	33.3245428	-	0.1742177
	45.3095930		1.3596463	
listed_inAnimation, Buddy, Kids	18.2787393	33.6276422	0.5435629	0.5868513
listed_inAnimation, Comedy	-	26.3526025	-	0.6077268
	13.5311913		0.5134670	
listed_inAnimation, Comedy, Coming of Age	5.2756142	24.7872034	0.2128362	0.8314938
listed_inAnimation, Comedy, Crime	33.3916514	33.3293402	1.0018696	0.3166248
listed_inAnimation, Comedy, Dance	7.8906390	28.7934697	0.2740427	0.7841028
listed_inAnimation, Comedy, Disaster	42.7277322	33.3097661	1.2827389	0.1998511
listed_inAnimation, Comedy, Drama	21.3516668	27.2163241	0.7845169	0.4329041
listed_inAnimation, Comedy, Family	6.3091328	23.6223173	0.2670836	0.7894544
listed_inAnimation, Comedy, Fantasy	11.1654865	28.8597547	0.3868878	0.6989134
listed_inAnimation, Comedy, Kids	7.4861702	26.3046594	0.2845948	0.7760076
listed_inAnimation, Comedy, Parody	1.7870546	33.3312744	0.0536150	0.9572516
listed_inAnimation, Coming of Age, Family	15.8630147	27.2241214	0.5826823	0.5602254
listed_inAnimation, Coming of Age, Fantasy	25.3674751	27.1349138	0.9348648	0.3500611
listed_inAnimation, Crime, Family	-	33.3076032	-	0.2988035
	34.6228899		1.0394891	
listed_inAnimation, Dance, Family	-	33.3065910	-	0.3339528
	32.1942587		0.9666032	
listed_inAnimation, Documentary, Family	43.8568438	33.2724725	1.3181120	0.1877379
listed_inAnimation, Drama	-	33.3092484	-	0.3055805
	34.1423922		1.0250124	
listed_inAnimation, Drama, Family	19.8053471	27.2030371	0.7280565	0.4667324
listed_inAnimation, Drama, Fantasy	35.0783544	33.3129929	1.0529932	0.2925731
listed_inAnimation, Drama, Kids	33.8576078	33.3092484	1.0164627	0.3096304
listed_inAnimation, Family	8.8236612	23.9789090	0.3679759	0.7129613
listed_inAnimation, Family, Fantasy	17.7977671	23.9471394	0.7432106	0.4575113
listed_inAnimation, Family, Historical	17.2024053	33.3662068	0.5155637	0.6062617
listed_inAnimation, Family, Kids	-	24.3715377	-	0.8226101
	5.4651278		0.2242422	
listed_inAnimation, Family, Music	-	33.3375896	-	0.9014224
	4.1302942		0.1238930	
listed_inAnimation, Family, Musical	0.8989096	26.4051356	0.0340430	0.9728490
listed_inAnimation, Family, Mystery	31.2600964	33.3276853	0.9379618	0.3484679
listed_inAnimation, Family, Parody	7.9701057	26.3402694	0.3025825	0.7622647
listed_inAnimation, Family, Romance	-	33.3342602	-	0.1387338
	49.3876021		1.4815869	
listed_inAnimation, Family, Sports	1.6225943	26.4078685	0.0614436	0.9510170
listed_inAnimation, Fantasy	46.7623113	29.1102637	1.6063857	0.1084734
listed_inAnimation, Fantasy, Kids	26.9068592	25.2428057	1.0659219	0.2866905
listed_inAnimation, Fantasy, Musical	21.1688801	33.2096829	0.6374310	0.5239755
listed_inAnimation, Kids	14.4148140	24.3958381	0.5908718	0.5547265
listed_inAnimation, Kids, Medical	31.2165155	28.9741530	1.0773918	0.2815391
listed_inAnimation, Kids, Music	17.4086148	29.2423573	0.5953219	0.5517496
listed_inAnimation, Kids, Musical	37.2902607	26.5967161	1.4020626	0.1611759

term	estimate	std.error	statistic	p.value
listed_inAnimation, Kids, Parody	-	33.2080390	-	0.9697462
	1.2597511		0.0379351	
listed_inAnimation, Kids, Police/Cop	17.6683660	33.6272865	0.5254175	0.5993978
listed_inAnimation, Kids, Sports	17.7787393	29.2425421	0.6079752	0.5433282
listed_inAnimation, Kids, Western	17.5189881	29.2430572	0.5990820	0.5492404
listed_inAnimation, Science Fiction	18.6812279	33.6397090	0.5553326	0.5787790
listed_inAnimation, Superhero	18.1542918	33.3561784	0.5442557	0.5863747
listed_inAnthology, Comedy, Coming of Age	42.3935273	28.7945751	1.4722748	0.1412299
listed_inAnthology, Comedy, Family	-	33.2089997	-	0.9978172
	0.0908710		0.0027363	
listed_inAnthology, Comedy, Kids	18.2841673	33.3561370	0.5481500	0.5836990
listed_inAnthology, Documentary	60.8640903	33.4932403	1.8172052	0.0694550
listed_inAnthology, Docuseries	0.7179482	28.0342127	0.0256097	0.9795732
listed_inAnthology, Docuseries, Family	-	28.9011213	-	0.9611596
	1.4077587		0.0487095	
listed_inAnthology, Family, Lifestyle	-	33.2089997	-	0.9978172
	0.0908710		0.0027363	
listed_inAnthology, Science Fiction	0.1688801	33.2096829	0.0050853	0.9959435
listed_inBiographical, Comedy, Drama	31.9411796	33.5835041	0.9510973	0.3417618
listed_inBiographical, Comedy, Family	42.3136423	33.3458604	1.2689324	0.2047310
listed_inBiographical, Coming of Age, Drama	86.8907876	33.3473593	2.6056272	0.0092932
listed_inBiographical, Coming of Age, Family	123.7924607	33.3609420	3.7107004	0.0002169
listed_inBiographical, Documentary	46.0519027	25.0593327	1.8377146	0.0663714
listed_inBiographical, Documentary, Music	64.5472026	26.1376761	2.4695081	0.0136794
listed_inBiographical, Drama	65.0998154	27.1772953	2.3953751	0.0167681
listed_inBiographical, Drama, Family	63.3657181	28.8944019	2.1930102	0.0285134
listed_inBiographical, Drama, Historical	69.1757347	28.9475968	2.3896883	0.0170286
listed_inBiographical, Drama, Music	47.4606818	33.5814964	1.4132986	0.1578481
listed_inBiographical, Drama, Sports	72.8911452	26.3390986	2.7674123	0.0057442
listed_inBiographical, Family, Historical	128.5327095	33.3653332	3.8522831	0.0001237
listed_inBiographical, Family, Sports	68.6700651	28.8612398	2.3793179	0.0175128
listed_inBiographical, Historical, Musical	85.4606818	33.5814964	2.5448741	0.0110661
listed_inBiographical, Sports	78.2866967	28.7980268	2.7184743	0.0066605
listed_inBuddy, Comedy	41.7487045	28.9188602	1.4436497	0.1491194
listed_inBuddy, Comedy, Coming of Age	45.9542393	24.7865764	1.8539970	0.0640045
listed_inBuddy, Comedy, Docuseries	4.6181745	33.3320809	0.1385504	0.8898305
listed_inBuddy, Comedy, Drama	49.8202826	33.3242971	1.4950138	0.1351948
listed_inBuddy, Comedy, Family	0.0078012	33.3314292	0.0002341	0.9998133
listed_inBuddy, Comedy, Romance	46.9240156	33.3501687	1.4070098	0.1597041
listed_inBuddy, Coming of Age, Drama	87.0908710	33.2089997	2.6225081	0.0088479
listed_inBuddy, Coming of Age, Family	-	33.2772452	-	0.8006920
	8.4029073		0.2525121	
listed_inBuddy, Docuseries	-	33.3311938	-	0.9678714
	1.3428209		0.0402872	
listed_inBuddy, Docuseries, Lifestyle	-	33.3312970	-	0.9793647
	0.8623232		0.0258713	

term	estimate	std.error	statistic	p.value
listed_inBuddy, Family, Sports	84.8311199	33.2096829	2.5544092	0.0107693
listed_inComedy	23.1426327	24.9825466	0.9263520	0.3544642
listed_inComedy, Coming of Age	72.0949740	24.6311424	2.9269846	0.0034924
listed_inComedy, Coming of Age, Crime	48.3972427	28.8837728	1.6755859	0.0941007
listed_inComedy, Coming of Age, Drama	38.2345394	24.4108775	1.5662911	0.1175653
listed_inComedy, Coming of Age, Family	25.3658460	24.2648462	1.0453743	0.2960774
listed_inComedy, Coming of Age, Fantasy	56.6257875	24.7618514	2.2868156	0.0223940
listed_inComedy, Coming of Age, Kids	-	25.1084442	-	0.9970279
	0.0935510		0.0037259	
listed_inComedy, Coming of Age, Medical	19.1542918	33.3561784	0.5742352	0.5659249
listed_inComedy, Coming of Age, Music	69.1471644	25.3634408	2.7262533	0.0065065
listed_inComedy, Coming of Age, Romance	10.1068963	28.8740390	0.3500340	0.7263795
listed_inComedy, Coming of Age, Science Fiction	84.0067050	25.7318554	3.2646968	0.0011292
listed_inComedy, Coming of Age, Sports	82.8311199	33.2096829	2.4941858	0.0127691
listed_inComedy, Crime, Family	48.8972427	28.8837728	1.6928967	0.0907555
listed_inComedy, Crime, Fantasy	83.5713688	33.2105368	2.5164113	0.0119957
listed_inComedy, Crime, Science Fiction	44.0522118	33.3567202	1.3206398	0.1868935
listed_inComedy, Docuseries, Family	18.9335452	33.3565677	0.5676107	0.5704140
listed_inComedy, Drama	47.9540415	33.6081490	1.4268576	0.1539019
listed_inComedy, Drama, Family	48.8944470	26.3880187	1.8529033	0.0641612
listed_inComedy, Drama, Romance	35.1728464	28.9559800	1.2147006	0.2247384
listed_inComedy, Drama, Sports	14.9192920	27.2283096	0.5479331	0.5838479
listed_inComedy, Drama, Thriller	8.9549613	33.4948123	0.2673537	0.7892465
listed_inComedy, Family	36.9345271	24.1699479	1.5281178	0.1267679
listed_inComedy, Family, Fantasy	53.5406799	24.1410145	2.2178306	0.0267681
listed_inComedy, Family, Game Show / Competition	-	33.2087222	-	0.9946975
	0.2207466		0.0066472	
listed_inComedy, Family, Kids	19.4782946	26.2915123	0.7408587	0.4589357
listed_inComedy, Family, Lifestyle	-	33.3311938	-	0.9678714
	1.3428209		0.0402872	
listed_inComedy, Family, Music	34.7576343	26.3185369	1.3206522	0.1868893
listed_inComedy, Family, Musical	27.4522068	28.8509903	0.9515170	0.3415490
listed_inComedy, Family, Romance	75.8580280	27.2421309	2.7845850	0.0054506
listed_inComedy, Family, Romantic Comedy	43.7015896	33.3469076	1.3105140	0.1902929
listed_inComedy, Family, Science Fiction	44.4719175	25.1823241	1.7659973	0.0776709
listed_inComedy, Family, Sports	47.7185349	25.8699933	1.8445515	0.0653689
listed_inComedy, Family, Variety	-	33.3697428	-	0.9270162
	3.0573459		0.0916203	
listed_inComedy, Family, Western	49.5912163	28.9130681	1.7151835	0.0865904
listed_inComedy, Fantasy	64.0685830	25.1926017	2.5431507	0.0111205
listed_inComedy, Fantasy, Kids	41.2595454	27.2170402	1.5159453	0.1298176
listed_inComedy, Fantasy, Romance	102.9816587	33.3566166	3.0872933	0.0020699
listed_inComedy, Kids	16.6098592	29.2463872	0.5679286	0.5701982
listed_inComedy, Music	49.3381055	33.3115745	1.4811100	0.1388608
listed_inComedy, Musical, Romance	101.0522118	33.3567202	3.0294409	0.0025067
listed_inComedy, Mystery	82.6103733	33.2081458	2.4876539	0.0130047

term	estimate	std.error	statistic	p.value
listed_inComedy, Parody, Science Fiction	-	33.3311938	-	0.9678714
	1.3428209		0.0402872	
listed_inComedy, Reality	20.5439185	33.3561818	0.6158954	0.5380897
listed_inComedy, Romance, Romantic Comedy	51.3916514	33.3293402	1.5419343	0.1233743
listed_inComedy, Science Fiction	67.0179637	27.1333662	2.4699465	0.0136627
listed_inComedy, Sports	42.8721491	33.3324501	1.2861986	0.1986418
listed_inComedy, Variety	22.0390045	33.2093200	0.6636391	0.5070590
listed_inComing of Age, Dance, Drama	81.7012443	33.2100885	2.4601333	0.0140399
listed_inComing of Age, Dance, Sports	83.3506222	33.2084874	2.5099193	0.0122172
listed_inComing of Age, Documentary, Drama	21.9091290	33.2089997	0.6597347	0.5095608
listed_inComing of Age, Docuseries, Family	-	33.2084874	-	0.9915778
	0.3506222		0.0105582	
listed_inComing of Age, Drama	67.5603524	26.2647422	2.5722831	0.0102320
listed_inComing of Age, Drama, Family	70.8845130	25.5023698	2.7795265	0.0055357
listed_inComing of Age, Drama, Fantasy	60.1568211	28.7987967	2.0888658	0.0369469
listed_inComing of Age, Drama, Music	64.3175387	25.1063518	2.5618035	0.0105441
listed_inComing of Age, Drama, Musical	27.1698924	28.7923948	0.9436482	0.3455546
listed_inComing of Age, Drama, Romance	64.7034683	27.1962116	2.3791353	0.0175214
listed_inComing of Age, Drama, Science Fiction	-	33.4938846	-	0.9605900
	1.6554120		0.0494243	
listed_inComing of Age, Drama, Sports	88.0233190	25.7258669	3.4215881	0.0006452
listed_inComing of Age, Family, Musical	80.1173589	33.3083407	2.4053242	0.0163208
listed_inComing of Age, Fantasy	73.2082776	25.1548396	2.9103059	0.0036828
listed_inComing of Age, Fantasy, Musical	102.2234027	25.7234053	3.9739452	0.0000753
listed_inComing of Age, Fantasy, Mystery	-	33.3314292	-	0.9762577
	0.9921988		0.0297677	
listed_inComing of Age, Music	93.1688801	33.2096829	2.8054733	0.0051119
listed_inComing of Age, Music, Musical	96.1103733	28.7591533	3.3419055	0.0008598
listed_inComing of Age, Musical	111.3506222	33.2084874	3.3530772	0.0008261
listed_inComing of Age, Musical, Romance	97.2207466	33.2087222	2.9275666	0.0034859
listed_inComing of Age, Musical, Science Fiction	95.7792534	33.2087222	2.8841596	0.0040003
listed_inComing of Age, Sports	87.5259332	28.7603181	3.0432881	0.0023951
listed_inConcert Film, Documentary, Music	45.1822967	28.8745592	1.5647788	0.1179196
listed_inConcert Film, Family, Fantasy	81.5273035	33.3311558	2.4459789	0.0146002
listed_inConcert Film, Music	68.0568824	27.1976264	2.5023096	0.0124814
listed_inConcert Film, Music, Musical	45.6571791	33.3311938	1.3698033	0.1710251
listed_inCrime, Drama	62.5905574	33.5811000	1.8638626	0.0626046
listed_inCrime, Drama, Family	88.6232353	33.2223826	2.6675761	0.0077512
listed_inCrime, Drama, Mystery	58.5472508	33.4034915	1.7527285	0.0799244
listed_inDance, Music, Musical	81.7342147	33.4933379	2.4403126	0.0148300
listed_inDocumentary	61.8125651	24.1126483	2.5634913	0.0104932
listed_inDocumentary, Family	38.7489790	24.5686425	1.5771722	0.1150406
listed_inDocumentary, Family, Historical	68.3712854	28.9255971	2.3636949	0.0182651
listed_inDocumentary, Family, Reality	39.2675524	33.3312074	1.1781017	0.2390083
listed_inDocumentary, Family, Variety	41.6438984	33.3899655	1.2471980	0.2125877
listed_inDocumentary, Historical	47.4164872	24.1775623	1.9611773	0.0501078

term	estimate	std.error	statistic	p.value
listed_inDocumentary, Music	102.7342147	33.4933379	3.0673030	0.0022123
listed_inDocumentary, Sports	48.2646641	28.8608600	1.6723224	0.0947423
listed_inDocuseries	-	25.8263148	-	0.9573673
	1.3809243		0.0534697	
listed_inDocuseries, Family	-	25.8816771	-	0.9586312
	1.3428209		0.0518831	
listed_inDocuseries, Family, Game Show / Competition	5.8701244	33.2079750	0.1767685	0.8597224
listed_inDocuseries, Family, Lifestyle	-	33.3312074	-	0.9824720
	0.7324476		0.0219748	
listed_inDocuseries, Family, Medical	-	33.3311558	-	0.9647658
	1.4726965		0.0441838	
listed_inDocuseries, Historical	-	24.8011736	-	0.9404400
	1.8534863		0.0747338	
listed_inDocuseries, Historical, Music	6.7624510	38.5480185	0.1754293	0.8607743
listed_inDocuseries, Historical, Reality	-	33.4933379	-	0.9223425
	3.2657853		0.0975055	
listed_inDocuseries, Kids	-	33.2087222	-	0.9946975
	0.2207466		0.0066472	
listed_inDocuseries, Reality	-	27.1812847	-	0.9614191
	1.3151310		0.0483837	
listed_inDocuseries, Reality, Survival	-	27.3425953	-	0.9907383
	0.3174669		0.0116107	
listed_inDocuseries, Science Fiction	-	33.3312970	-	0.9554529
	1.8623232		0.0558731	
listed_inDrama	-	27.2217306	-	0.2691130
	30.0979808		1.1056601	
listed_inDrama, Family	69.5949443	28.8668087	2.4108984	0.0160748
listed_inDrama, Family, Fantasy	52.3848005	26.3722591	1.9863600	0.0472387
listed_inDrama, Family, Historical	37.4750183	33.4117841	1.1216108	0.2622703
listed_inDrama, Family, Musical	85.9297922	33.3393283	2.5774302	0.0100818
listed_inDrama, Family, Mystery	39.2414098	33.3537594	1.1765213	0.2396386
listed_inDrama, Fantasy	67.3507948	28.7990999	2.3386424	0.0195305
listed_inDrama, Fantasy, Mystery	-	33.4932403	-	0.9254214
	3.1359097		0.0936281	
listed_inDrama, Historical	39.3055835	29.0924509	1.3510578	0.1769520
listed_inDrama, Historical, Sports	69.9825224	28.9058029	2.4210544	0.0156350
listed_inDrama, Music	85.1607218	28.8665770	2.9501496	0.0032428
listed_inDrama, Music, Romance	45.0390045	33.2093200	1.3562158	0.1753061
listed_inDrama, Mystery	80.7012443	33.2100885	2.4300220	0.0152555
listed_inDrama, Parody, Science Fiction	-	33.3599314	-	0.3552088
	30.8552378		0.9249191	
listed_inDrama, Romance	44.6034194	33.6008941	1.3274474	0.1846334
listed_inDrama, Sports	82.8711368	28.7951711	2.8779526	0.0040793
listed_inFamily, Fantasy	84.8389211	33.3345328	2.5450761	0.0110597
listed_inFamily, Fantasy, Musical	90.5971355	26.3768520	3.4347213	0.0006151

term	estimate	std.error	statistic	p.value
listed_inFamily, Fantasy, Mystery	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inFamily, Fantasy, Romance	43.7534561	33.3795229	1.3107873	0.1902005
listed_inFamily, Game Show / Competition	3.4376471	25.7415546	0.1335447	0.8937868
listed_inFamily, Game Show / Competition, Reality	0.0390045	33.2093200	0.0011745	0.9990631
listed_inFamily, Kids	18.5439185	33.3561818	0.5559365	0.5783662
listed_inFamily, Lifestyle, Reality	50.2675524	28.9011651	1.7392916	0.0822605
listed_inFamily, Music, Variety	32.6039423	28.7607125	1.1336278	0.2571953
listed_inFamily, Variety	36.5011609	28.9240722	1.2619648	0.2072263
listed_inFantasy	27.5171446	28.8509071	0.9537705	0.3404073
listed_inFantasy, Kids, Mystery	18.3360338	33.3612209	0.5496212	0.5826897
listed_inFantasy, Musical	50.9828263	27.1285989	1.8793019	0.0604649
listed_inFantasy, Musical, Thriller	8.7792534	33.2087222	0.2643659	0.7915470
listed_inFantasy, Variety	49.0390045	33.2093200	1.4766639	0.1400491
listed_inGame Show / Competition, Sports, Variety	-	33.2082952	-	0.9635124
	1.5195023		0.0457567	
listed_inHistorical, Musical	37.6571791	28.9011494	1.3029648	0.1928568
listed_inKids	-	33.2121378	-	0.9803487
	0.8182579		0.0246373	
listed_inKids, Music	18.1942992	29.2439738	0.6221555	0.5339673
listed_inLifestyle	-	33.4932403	-	0.9491637
	2.1359097		0.0637714	
listed_inMovies	7.5439335	27.1580059	0.2777794	0.7812334
listed_inMusic, Reality	-	33.4934778	-	0.9192649
	3.3956609		0.1013827	
listed_inMusical	38.7870546	33.3312744	1.1636835	0.2448021
listed_inMusical, Romance, Western	137.3451428	33.4144197	4.1103555	0.0000424
listed_inReality	-	26.4466776	-	0.9698711
	0.9991165		0.0377785	
listed_inScience Fiction	18.4779674	28.7953389	0.6416999	0.5212004
listed_inSeries	-	27.6928409	-	0.9450611
	1.9087521		0.0689258	
listed_inSports	87.2597511	33.2080390	2.6276695	0.0087157
listed_inTalk Show	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inVariety	22.5195023	33.2082952	0.6781288	0.4978312

```
glance(model_multi)
```

r.squaredadj	r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	cdf.residual	nobs
0.8049636	0.7456276	23.481571	13.56619	0	338	-	13561.8115356	7861.2587.7	1111	1450	6440.904

```
model_multi1 <- lm(duration_num ~ release_year + rating, data = disney)
tidy(model_multi)
```

term	estimate	std.error	statistic	p.value
(Intercept)	256.7160698	82.0478244	3.1288589	0.0018006
release_year	- 0.1298756	0.0376554	- 3.4490523	0.0005837
ratingG	44.9567097	19.5768813	2.2964184	0.0218374
ratingPG	58.8122928	19.5623930	3.0063956	0.0027030
ratingPG-13	80.1718936	19.9736079	4.0138914	0.0000637
ratingTV-14	9.8983607	19.5955230	0.5051338	0.6135651
ratingTV-G	6.5935709	19.5786190	0.3367741	0.7363508
ratingTV-PG	7.9753963	19.4143681	0.4107987	0.6812994
ratingTV-Y	- 10.9059150	20.1502627	- 0.5412294	0.5884580
ratingTV-Y7	- 12.4308453	19.6940199	- 0.6311990	0.5280403
ratingTV-Y7-FV	8.6294797	21.0412526	0.4101220	0.6817956
listed_inAction-Adventure, Animals & Nature, Comedy	52.5733934	33.3432004	1.5767351	0.1151412
listed_inAction-Adventure, Animals & Nature, Documentary	57.4323556	25.2020986	2.2788719	0.0228638
listed_inAction-Adventure, Animals & Nature, Docuseries	0.6930826	24.8341052	0.0279085	0.9777401
listed_inAction-Adventure, Animals & Nature, Drama	51.9560387	25.8496268	2.0099338	0.0446796
listed_inAction-Adventure, Animals & Nature, Family	46.6872962	24.6522841	1.8938325	0.0585068
listed_inAction-Adventure, Animation	49.7159207	26.3083986	1.8897357	0.0590535
listed_inAction-Adventure, Animation, Anime	14.1274203	26.3866838	0.5353996	0.5924807
listed_inAction-Adventure, Animation, Anthology	11.2891264	25.8005415	0.4375539	0.6617948
listed_inAction-Adventure, Animation, Buddy	31.1302192	25.0306155	1.2436857	0.2138776
listed_inAction-Adventure, Animation, Comedy	26.8683769	23.7187774	1.1327893	0.2575472
listed_inAction-Adventure, Animation, Coming of Age	45.1274428	24.7082250	1.8264138	0.0680562
listed_inAction-Adventure, Animation, Family	40.6280493	23.8968926	1.7001394	0.0893846
listed_inAction-Adventure, Animation, Fantasy	22.1605221	24.3581387	0.9097790	0.3631364
listed_inAction-Adventure, Animation, Kids	20.9852799	23.8950508	0.8782270	0.3800104
listed_inAction-Adventure, Animation, Musical	41.8897994	28.7920758	1.4549072	0.1459774
listed_inAction-Adventure, Animation, Science Fiction	16.7941762	25.8676240	0.6492354	0.5163204
listed_inAction-Adventure, Animation, Superhero	30.9033826	29.1331713	1.0607627	0.2890283
listed_inAction-Adventure, Biographical, Drama	45.4435178	33.3445092	1.3628486	0.1732064
listed_inAction-Adventure, Buddy, Comedy	- 1.2129454	33.3312744	- 0.0363906	0.9709774

term	estimate	std.error	statistic	p.value
listed_inAction-Adventure, Buddy, Science Fiction	- 3.1359097	33.4932403	- 0.0936281	0.9254214
listed_inAction-Adventure, Comedy	90.8311199	33.2096829	2.7350794	0.0063358
listed_inAction-Adventure, Comedy, Coming of Age	41.7651564	24.3007175	1.7186800	0.0859512
listed_inAction-Adventure, Comedy, Crime	37.1383312	28.9142594	1.2844296	0.1992594
listed_inAction-Adventure, Comedy, Family	50.5400022	24.0272140	2.1034483	0.0356510
listed_inAction-Adventure, Comedy, Fantasy	60.8708424	26.3360597	2.3113117	0.0209980
listed_inAction-Adventure, Comedy, Kids	52.6909004	25.7925074	2.0428762	0.0413003
listed_inAction-Adventure, Comedy, Science Fiction	56.8641265	25.2786607	2.2494913	0.0246764
listed_inAction-Adventure, Comedy, Spy/Espionage	41.8721491	33.3324501	1.2561978	0.2093084
listed_inAction-Adventure, Coming of Age	89.5713688	33.2105368	2.6970768	0.0071010
listed_inAction-Adventure, Coming of Age, Drama	52.1848422	26.4100797	1.9759441	0.0484082
listed_inAction-Adventure, Coming of Age, Family	56.3801610	24.8755202	2.2664917	0.0236129
listed_inAction-Adventure, Coming of Age, Fantasy	86.8389211	33.3345328	2.6050739	0.0093082
listed_inAction-Adventure, Crime, Family	88.3841473	33.3945181	2.6466664	0.0082439
listed_inAction-Adventure, Crime, Fantasy	42.8202826	33.3242971	1.2849568	0.1990752
listed_inAction-Adventure, Crime, Science Fiction	60.2009307	33.5824158	1.7926325	0.0733037
listed_inAction-Adventure, Disaster, Drama	43.9411796	33.5835041	1.3084156	0.1910030
listed_inAction-Adventure, Documentary	42.2597511	33.2080390	1.2725759	0.2034348
listed_inAction-Adventure, Documentary, Historical	91.5273035	33.3311558	2.7459985	0.0061301
listed_inAction-Adventure, Documentary, Sports	98.3974279	33.3311604	2.9521153	0.0032224
listed_inAction-Adventure, Documentary, Superhero	39.8779257	33.3316039	1.1963998	0.2317958
listed_inAction-Adventure, Docuseries, Family	- 1.9535187	27.2754827	- 0.0716218	0.9429158
listed_inAction-Adventure, Docuseries, Fantasy	- 3.1359097	33.4932403	- 0.0936281	0.9254214
listed_inAction-Adventure, Docuseries, Historical	- 1.4726965	28.9011055	- 0.0509564	0.9593694
listed_inAction-Adventure, Docuseries, Lifestyle	- 1.5255364	33.4936600	- 0.0455470	0.9636795
listed_inAction-Adventure, Docuseries, Reality	- 3.1359097	33.4932403	- 0.0936281	0.9254214
listed_inAction-Adventure, Drama, Family	42.8833317	33.3770522	1.2848148	0.1991248
listed_inAction-Adventure, Drama, Historical	50.4216773	33.5861873	1.5012623	0.1335719
listed_inAction-Adventure, Family	18.1682639	27.2846674	0.6658782	0.5056272
listed_inAction-Adventure, Family, Fantasy	62.8101740	24.3474911	2.5797391	0.0100151
listed_inAction-Adventure, Family, Game Show / Competition	- 3.1359097	33.4932403	- 0.0936281	0.9254214

term	estimate	std.error	statistic	p.value
listed_inAction-Adventure, Family, Historical	32.3711001	28.9604751	1.1177683	0.2639076
listed_inAction-Adventure, Family, Mystery	70.9613408	33.3432794	2.1282052	0.0335397
listed_inAction-Adventure, Family, Parody	30.1319002	33.3308101	0.9040254	0.3661779
listed_inAction-Adventure, Family, Science Fiction	46.1674212	24.4480011	1.8883925	0.0592336
listed_inAction-Adventure, Family, Survival	30.4302372	33.4780592	0.9089606	0.3635681
listed_inAction-Adventure, Family, Western	45.7015896	33.3469076	1.3704896	0.1708110
listed_inAction-Adventure, Fantasy	58.9991209	24.8769456	2.3716385	0.0178791
listed_inAction-Adventure, Fantasy, Musical	72.0410292	33.3264098	2.1616799	0.0308559
listed_inAction-Adventure, Fantasy, Science Fiction	67.2088929	25.3499147	2.6512473	0.0081336
listed_inAction-Adventure, Fantasy, Soap Opera / Melodrama	0.6181745	28.9021724	0.0213885	0.9829396
listed_inAction-Adventure, Fantasy, Superhero	46.9411796	27.5728554	1.7024417	0.0889523
listed_inAction-Adventure, Fantasy, Survival	48.7015896	33.3469076	1.4604530	0.1444484
listed_inAction-Adventure, Mystery, Thriller	73.4370869	28.9021070	2.5408904	0.0111922
listed_inAction-Adventure, Reality, Survival	-	33.3311558	-	0.9886875
	0.4726965		0.0141818	
listed_inAction-Adventure, Science Fiction	66.1721683	24.4817512	2.7029181	0.0069782
listed_inAction-Adventure, Science Fiction, Spy/Espionage	28.7727593	28.9976340	0.9922451	0.3212940
listed_inAction-Adventure, Science Fiction, Superhero	52.5917722	24.9681188	2.1063570	0.0353972
listed_inAction-Adventure, Science Fiction, Thriller	39.4953843	33.3692333	1.1835868	0.2368298
listed_inAction-Adventure, Superhero	-	33.4932403	-	0.9254214
	3.1359097		0.0936281	
listed_inAction-Adventure, Western	40.5733934	33.3432004	1.2168416	0.2239229
listed_inAnimals & Nature	-	33.3312744	-	0.9949037
	0.2129454		0.0063888	
listed_inAnimals & Nature, Animation, Family	-	33.3064125	-	0.3097888
	33.8436365		1.0161297	
listed_inAnimals & Nature, Animation, Kids	-	33.2404100	-	0.9744429
	1.0651471		0.0320437	
listed_inAnimals & Nature, Anthology	0.1688801	33.2096829	0.0050853	0.9959435
listed_inAnimals & Nature, Biographical, Documentary	54.1265277	27.3148807	1.9815766	0.0477728
listed_inAnimals & Nature, Biographical, Drama	75.1632521	28.8111872	2.6088218	0.0092075
listed_inAnimals & Nature, Comedy	46.6514025	33.3280404	1.3997643	0.1618631
listed_inAnimals & Nature, Comedy, Coming of Age	85.3311199	28.7609282	2.9669112	0.0030725
listed_inAnimals & Nature, Comedy, Family	44.6258521	25.1959726	1.7711502	0.0768098
listed_inAnimals & Nature, Coming of Age, Drama	74.3296019	28.8606342	2.5754667	0.0101389
listed_inAnimals & Nature, Coming of Age, Family	85.4414932	33.2110278	2.5726844	0.0102202

term	estimate	std.error	statistic	p.value
listed_inAnimals & Nature, Documentary	45.1014933	23.9935138	1.8797369	0.0604055
listed_inAnimals & Nature, Documentary, Family	42.5968435	23.8821880	1.7836240	0.0747577
listed_inAnimals & Nature, Documentary, Medical	39.8250857	33.4952062	1.1889787	0.2347021
listed_inAnimals & Nature, Documentary, Survival	55.7342147	33.4933379	1.6640388	0.0963867
listed_inAnimals & Nature, Docuseries	-	24.7930762	-	0.9390485
	1.8962520		0.0764831	
listed_inAnimals & Nature, Docuseries, Family	-	23.8969174	-	0.9821692
	0.5342033		0.0223545	
listed_inAnimals & Nature, Docuseries, Kids	-	27.1149243	-	0.9909587
	0.3073303		0.0113344	
listed_inAnimals & Nature, Docuseries, Medical	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAnimals & Nature, Docuseries, Reality	-	25.9018327	-	0.9350281
	2.1120067		0.0815389	
listed_inAnimals & Nature, Docuseries, Survival	-	28.9015154	-	0.9432693
	2.0571366		0.0711775	
listed_inAnimals & Nature, Docuseries, Travel	-	33.3314292	-	0.9523501
	1.9921988		0.0597694	
listed_inAnimals & Nature, Drama, Family	63.8872202	25.4891510	2.5064476	0.0123371
listed_inAnimals & Nature, Family	3.0912163	33.3415289	0.0927137	0.9261477
listed_inAnimals & Nature, Family, Reality	41.0078012	33.3314292	1.2303043	0.2188436
listed_inAnimals & Nature, Family, Sports	80.2932763	33.3793247	2.4054794	0.0163139
listed_inAnimals & Nature, Kids, Music	-	33.2082952	-	0.9884582
	0.4804977		0.0144692	
listed_inAnimals & Nature, Kids, Variety	0.7143156	28.7596772	0.0248374	0.9801891
listed_inAnimals & Nature, Reality	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAnimation	3.1509257	29.0450521	0.1084841	0.9136313
listed_inAnimation, Anthology, Comedy	30.3031581	27.1296518	1.1169756	0.2642463
listed_inAnimation, Anthology, Family	3.2381200	26.3254045	0.1230036	0.9021265
listed_inAnimation, Anthology, Fantasy	28.8186033	33.3145815	0.8650447	0.3872010
listed_inAnimation, Anthology, Kids	19.7592370	33.6287067	0.5875705	0.5569400
listed_inAnimation, Anthology, Science Fiction	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inAnimation, Buddy, Comedy	17.9752526	25.4417901	0.7065247	0.4800101
listed_inAnimation, Buddy, Coming of Age	33.5717272	28.8614987	1.1632011	0.2449977
listed_inAnimation, Buddy, Family	-	33.3245428	-	0.1742177
	45.3095930		1.3596463	
listed_inAnimation, Buddy, Kids	18.2787393	33.6276422	0.5435629	0.5868513
listed_inAnimation, Comedy	-	26.3526025	-	0.6077268
	13.5311913		0.5134670	
listed_inAnimation, Comedy, Coming of Age	5.2756142	24.7872034	0.2128362	0.8314938
listed_inAnimation, Comedy, Crime	33.3916514	33.3293402	1.0018696	0.3166248
listed_inAnimation, Comedy, Dance	7.8906390	28.7934697	0.2740427	0.7841028

term	estimate	std.error	statistic	p.value
listed_inAnimation, Comedy, Disaster	42.7277322	33.3097661	1.2827389	0.1998511
listed_inAnimation, Comedy, Drama	21.3516668	27.2163241	0.7845169	0.4329041
listed_inAnimation, Comedy, Family	6.3091328	23.6223173	0.2670836	0.7894544
listed_inAnimation, Comedy, Fantasy	11.1654865	28.8597547	0.3868878	0.6989134
listed_inAnimation, Comedy, Kids	7.4861702	26.3046594	0.2845948	0.7760076
listed_inAnimation, Comedy, Parody	1.7870546	33.3312744	0.0536150	0.9572516
listed_inAnimation, Coming of Age, Family	15.8630147	27.2241214	0.5826823	0.5602254
listed_inAnimation, Coming of Age, Fantasy	25.3674751	27.1349138	0.9348648	0.3500611
listed_inAnimation, Crime, Family	-	33.3076032	-	0.2988035
	34.6228899		1.0394891	
listed_inAnimation, Dance, Family	-	33.3065910	-	0.3339528
	32.1942587		0.9666032	
listed_inAnimation, Documentary, Family	43.8568438	33.2724725	1.3181120	0.1877379
listed_inAnimation, Drama	-	33.3092484	-	0.3055805
	34.1423922		1.0250124	
listed_inAnimation, Drama, Family	19.8053471	27.2030371	0.7280565	0.4667324
listed_inAnimation, Drama, Fantasy	35.0783544	33.3129929	1.0529932	0.2925731
listed_inAnimation, Drama, Kids	33.8576078	33.3092484	1.0164627	0.3096304
listed_inAnimation, Family	8.8236612	23.9789090	0.3679759	0.7129613
listed_inAnimation, Family, Fantasy	17.7977671	23.9471394	0.7432106	0.4575113
listed_inAnimation, Family, Historical	17.2024053	33.3662068	0.5155637	0.6062617
listed_inAnimation, Family, Kids	-	24.3715377	-	0.8226101
	5.4651278		0.2242422	
listed_inAnimation, Family, Music	-	33.3375896	-	0.9014224
	4.1302942		0.1238930	
listed_inAnimation, Family, Musical	0.8989096	26.4051356	0.0340430	0.9728490
listed_inAnimation, Family, Mystery	31.2600964	33.3276853	0.9379618	0.3484679
listed_inAnimation, Family, Parody	7.9701057	26.3402694	0.3025825	0.7622647
listed_inAnimation, Family, Romance	-	33.3342602	-	0.1387338
	49.3876021		1.4815869	
listed_inAnimation, Family, Sports	1.6225943	26.4078685	0.0614436	0.9510170
listed_inAnimation, Fantasy	46.7623113	29.1102637	1.6063857	0.1084734
listed_inAnimation, Fantasy, Kids	26.9068592	25.2428057	1.0659219	0.2866905
listed_inAnimation, Fantasy, Musical	21.1688801	33.2096829	0.6374310	0.5239755
listed_inAnimation, Kids	14.4148140	24.3958381	0.5908718	0.5547265
listed_inAnimation, Kids, Medical	31.2165155	28.9741530	1.0773918	0.2815391
listed_inAnimation, Kids, Music	17.4086148	29.2423573	0.5953219	0.5517496
listed_inAnimation, Kids, Musical	37.2902607	26.5967161	1.4020626	0.1611759
listed_inAnimation, Kids, Parody	-	33.2080390	-	0.9697462
	1.2597511		0.0379351	
listed_inAnimation, Kids, Police/Cop	17.6683660	33.6272865	0.5254175	0.5993978
listed_inAnimation, Kids, Sports	17.7787393	29.2425421	0.6079752	0.5433282
listed_inAnimation, Kids, Western	17.5189881	29.2430572	0.5990820	0.5492404
listed_inAnimation, Science Fiction	18.6812279	33.6397090	0.5553326	0.5787790
listed_inAnimation, Superhero	18.1542918	33.3561784	0.5442557	0.5863747
listed_inAnthology, Comedy, Coming of Age	42.3935273	28.7945751	1.4722748	0.1412299

term	estimate	std.error	statistic	p.value
listed_inAnthology, Comedy, Family	-	33.2089997	-	0.9978172
	0.0908710		0.0027363	
listed_inAnthology, Comedy, Kids	18.2841673	33.3561370	0.5481500	0.5836990
listed_inAnthology, Documentary	60.8640903	33.4932403	1.8172052	0.0694550
listed_inAnthology, Docuseries	0.7179482	28.0342127	0.0256097	0.9795732
listed_inAnthology, Docuseries, Family	-	28.9011213	-	0.9611596
	1.4077587		0.0487095	
listed_inAnthology, Family, Lifestyle	-	33.2089997	-	0.9978172
	0.0908710		0.0027363	
listed_inAnthology, Science Fiction	0.1688801	33.2096829	0.0050853	0.9959435
listed_inBiographical, Comedy, Drama	31.9411796	33.5835041	0.9510973	0.3417618
listed_inBiographical, Comedy, Family	42.3136423	33.3458604	1.2689324	0.2047310
listed_inBiographical, Coming of Age, Drama	86.8907876	33.3473593	2.6056272	0.0092932
listed_inBiographical, Coming of Age, Family	123.7924607	33.3609420	3.7107004	0.0002169
listed_inBiographical, Documentary	46.0519027	25.0593327	1.8377146	0.0663714
listed_inBiographical, Documentary, Music	64.5472026	26.1376761	2.4695081	0.0136794
listed_inBiographical, Drama	65.0998154	27.1772953	2.3953751	0.0167681
listed_inBiographical, Drama, Family	63.3657181	28.8944019	2.1930102	0.0285134
listed_inBiographical, Drama, Historical	69.1757347	28.9475968	2.3896883	0.0170286
listed_inBiographical, Drama, Music	47.4606818	33.5814964	1.4132986	0.1578481
listed_inBiographical, Drama, Sports	72.8911452	26.3390986	2.7674123	0.0057442
listed_inBiographical, Family, Historical	128.5327095	33.3653332	3.8522831	0.0001237
listed_inBiographical, Family, Sports	68.6700651	28.8612398	2.3793179	0.0175128
listed_inBiographical, Historical, Musical	85.4606818	33.5814964	2.5448741	0.0110661
listed_inBiographical, Sports	78.2866967	28.7980268	2.7184743	0.0066605
listed_inBuddy, Comedy	41.7487045	28.9188602	1.4436497	0.1491194
listed_inBuddy, Comedy, Coming of Age	45.9542393	24.7865764	1.8539970	0.0640045
listed_inBuddy, Comedy, Docuseries	4.6181745	33.3320809	0.1385504	0.8898305
listed_inBuddy, Comedy, Drama	49.8202826	33.3242971	1.4950138	0.1351948
listed_inBuddy, Comedy, Family	0.0078012	33.3314292	0.0002341	0.9998133
listed_inBuddy, Comedy, Romance	46.9240156	33.3501687	1.4070098	0.1597041
listed_inBuddy, Coming of Age, Drama	87.0908710	33.2089997	2.6225081	0.0088479
listed_inBuddy, Coming of Age, Family	-	33.2772452	-	0.8006920
	8.4029073		0.2525121	
listed_inBuddy, Docuseries	-	33.3311938	-	0.9678714
	1.3428209		0.0402872	
listed_inBuddy, Docuseries, Lifestyle	-	33.3312970	-	0.9793647
	0.8623232		0.0258713	
listed_inBuddy, Family, Sports	84.8311199	33.2096829	2.5544092	0.0107693
listed_inComedy	23.1426327	24.9825466	0.9263520	0.3544642
listed_inComedy, Coming of Age	72.0949740	24.6311424	2.9269846	0.0034924
listed_inComedy, Coming of Age, Crime	48.3972427	28.8837728	1.6755859	0.0941007
listed_inComedy, Coming of Age, Drama	38.2345394	24.4108775	1.5662911	0.1175653
listed_inComedy, Coming of Age, Family	25.3658460	24.2648462	1.0453743	0.2960774
listed_inComedy, Coming of Age, Fantasy	56.6257875	24.7618514	2.2868156	0.0223940

term	estimate	std.error	statistic	p.value
listed_inComedy, Coming of Age, Kids	-	25.1084442	-	0.9970279
	0.0935510		0.0037259	
listed_inComedy, Coming of Age, Medical	19.1542918	33.3561784	0.5742352	0.5659249
listed_inComedy, Coming of Age, Music	69.1471644	25.3634408	2.7262533	0.0065065
listed_inComedy, Coming of Age, Romance	10.1068963	28.8740390	0.3500340	0.7263795
listed_inComedy, Coming of Age, Science Fiction	84.0067050	25.7318554	3.2646968	0.0011292
listed_inComedy, Coming of Age, Sports	82.8311199	33.2096829	2.4941858	0.0127691
listed_inComedy, Crime, Family	48.8972427	28.8837728	1.6928967	0.0907555
listed_inComedy, Crime, Fantasy	83.5713688	33.2105368	2.5164113	0.0119957
listed_inComedy, Crime, Science Fiction	44.0522118	33.3567202	1.3206398	0.1868935
listed_inComedy, Docuseries, Family	18.9335452	33.3565677	0.5676107	0.5704140
listed_inComedy, Drama	47.9540415	33.6081490	1.4268576	0.1539019
listed_inComedy, Drama, Family	48.8944470	26.3880187	1.8529033	0.0641612
listed_inComedy, Drama, Romance	35.1728464	28.9559800	1.2147006	0.2247384
listed_inComedy, Drama, Sports	14.9192920	27.2283096	0.5479331	0.5838479
listed_inComedy, Drama, Thriller	8.9549613	33.4948123	0.2673537	0.7892465
listed_inComedy, Family	36.9345271	24.1699479	1.5281178	0.1267679
listed_inComedy, Family, Fantasy	53.5406799	24.1410145	2.2178306	0.0267681
listed_inComedy, Family, Game Show / Competition	-	33.2087222	-	0.9946975
	0.2207466		0.0066472	
listed_inComedy, Family, Kids	19.4782946	26.2915123	0.7408587	0.4589357
listed_inComedy, Family, Lifestyle	-	33.3311938	-	0.9678714
	1.3428209		0.0402872	
listed_inComedy, Family, Music	34.7576343	26.3185369	1.3206522	0.1868893
listed_inComedy, Family, Musical	27.4522068	28.8509903	0.9515170	0.3415490
listed_inComedy, Family, Romance	75.8580280	27.2421309	2.7845850	0.0054506
listed_inComedy, Family, Romantic Comedy	43.7015896	33.3469076	1.3105140	0.1902929
listed_inComedy, Family, Science Fiction	44.4719175	25.1823241	1.7659973	0.0776709
listed_inComedy, Family, Sports	47.7185349	25.8699933	1.8445515	0.0653689
listed_inComedy, Family, Variety	-	33.3697428	-	0.9270162
	3.0573459		0.0916203	
listed_inComedy, Family, Western	49.5912163	28.9130681	1.7151835	0.0865904
listed_inComedy, Fantasy	64.0685830	25.1926017	2.5431507	0.0111205
listed_inComedy, Fantasy, Kids	41.2595454	27.2170402	1.5159453	0.1298176
listed_inComedy, Fantasy, Romance	102.9816587	33.3566166	3.0872933	0.0020699
listed_inComedy, Kids	16.6098592	29.2463872	0.5679286	0.5701982
listed_inComedy, Music	49.3381055	33.3115745	1.4811100	0.1388608
listed_inComedy, Musical, Romance	101.0522118	33.3567202	3.0294409	0.0025067
listed_inComedy, Mystery	82.6103733	33.2081458	2.4876539	0.0130047
listed_inComedy, Parody, Science Fiction	-	33.3311938	-	0.9678714
	1.3428209		0.0402872	
listed_inComedy, Reality	20.5439185	33.3561818	0.6158954	0.5380897
listed_inComedy, Romance, Romantic Comedy	51.3916514	33.3293402	1.5419343	0.1233743
listed_inComedy, Science Fiction	67.0179637	27.1333662	2.4699465	0.0136627
listed_inComedy, Sports	42.8721491	33.3324501	1.2861986	0.1986418
listed_inComedy, Variety	22.0390045	33.2093200	0.6636391	0.5070590

term	estimate	std.error	statistic	p.value
listed_inComing of Age, Dance, Drama	81.7012443	33.2100885	2.4601333	0.0140399
listed_inComing of Age, Dance, Sports	83.3506222	33.2084874	2.5099193	0.0122172
listed_inComing of Age, Documentary, Drama	21.9091290	33.2089997	0.6597347	0.5095608
listed_inComing of Age, Docuseries, Family	-	33.2084874	-	0.9915778
	0.3506222		0.0105582	
listed_inComing of Age, Drama	67.5603524	26.2647422	2.5722831	0.0102320
listed_inComing of Age, Drama, Family	70.8845130	25.5023698	2.7795265	0.0055357
listed_inComing of Age, Drama, Fantasy	60.1568211	28.7987967	2.0888658	0.0369469
listed_inComing of Age, Drama, Music	64.3175387	25.1063518	2.5618035	0.0105441
listed_inComing of Age, Drama, Musical	27.1698924	28.7923948	0.9436482	0.3455546
listed_inComing of Age, Drama, Romance	64.7034683	27.1962116	2.3791353	0.0175214
listed_inComing of Age, Drama, Science Fiction	-	33.4938846	-	0.9605900
	1.6554120		0.0494243	
listed_inComing of Age, Drama, Sports	88.0233190	25.7258669	3.4215881	0.0006452
listed_inComing of Age, Family, Musical	80.1173589	33.3083407	2.4053242	0.0163208
listed_inComing of Age, Fantasy	73.2082776	25.1548396	2.9103059	0.0036828
listed_inComing of Age, Fantasy, Musical	102.2234027	25.7234053	3.9739452	0.0000753
listed_inComing of Age, Fantasy, Mystery	-	33.3314292	-	0.9762577
	0.9921988		0.0297677	
listed_inComing of Age, Music	93.1688801	33.2096829	2.8054733	0.0051119
listed_inComing of Age, Music, Musical	96.1103733	28.7591533	3.3419055	0.0008598
listed_inComing of Age, Musical	111.3506222	33.2084874	3.3530772	0.0008261
listed_inComing of Age, Musical, Romance	97.2207466	33.2087222	2.9275666	0.0034859
listed_inComing of Age, Musical, Science Fiction	95.7792534	33.2087222	2.8841596	0.0040003
listed_inComing of Age, Sports	87.5259332	28.7603181	3.0432881	0.0023951
listed_inConcert Film, Documentary, Music	45.1822967	28.8745592	1.5647788	0.1179196
listed_inConcert Film, Family, Fantasy	81.5273035	33.3311558	2.4459789	0.0146002
listed_inConcert Film, Music	68.0568824	27.1976264	2.5023096	0.0124814
listed_inConcert Film, Music, Musical	45.6571791	33.3311938	1.3698033	0.1710251
listed_inCrime, Drama	62.5905574	33.5811000	1.8638626	0.0626046
listed_inCrime, Drama, Family	88.6232353	33.2223826	2.6675761	0.0077512
listed_inCrime, Drama, Mystery	58.5472508	33.4034915	1.7527285	0.0799244
listed_inDance, Music, Musical	81.7342147	33.4933379	2.4403126	0.0148300
listed_inDocumentary	61.8125651	24.1126483	2.5634913	0.0104932
listed_inDocumentary, Family	38.7489790	24.5686425	1.5771722	0.1150406
listed_inDocumentary, Family, Historical	68.3712854	28.9255971	2.3636949	0.0182651
listed_inDocumentary, Family, Reality	39.2675524	33.3312074	1.1781017	0.2390083
listed_inDocumentary, Family, Variety	41.6438984	33.3899655	1.2471980	0.2125877
listed_inDocumentary, Historical	47.4164872	24.1775623	1.9611773	0.0501078
listed_inDocumentary, Music	102.7342147	33.4933379	3.0673030	0.0022123
listed_inDocumentary, Sports	48.2646641	28.8608600	1.6723224	0.0947423
listed_inDocuseries	-	25.8263148	-	0.9573673
	1.3809243		0.0534697	
listed_inDocuseries, Family	-	25.8816771	-	0.9586312
	1.3428209		0.0518831	

term	estimate	std.error	statistic	p.value
listed_inDocuseries, Family, Game Show / Competition	5.8701244	33.2079750	0.1767685	0.8597224
listed_inDocuseries, Family, Lifestyle	-	33.3312074	-	0.9824720
	0.7324476		0.0219748	
listed_inDocuseries, Family, Medical	-	33.3311558	-	0.9647658
	1.4726965		0.0441838	
listed_inDocuseries, Historical	-	24.8011736	-	0.9404400
	1.8534863		0.0747338	
listed_inDocuseries, Historical, Music	6.7624510	38.5480185	0.1754293	0.8607743
listed_inDocuseries, Historical, Reality	-	33.4933379	-	0.9223425
	3.2657853		0.0975055	
listed_inDocuseries, Kids	-	33.2087222	-	0.9946975
	0.2207466		0.0066472	
listed_inDocuseries, Reality	-	27.1812847	-	0.9614191
	1.3151310		0.0483837	
listed_inDocuseries, Reality, Survival	-	27.3425953	-	0.9907383
	0.3174669		0.0116107	
listed_inDocuseries, Science Fiction	-	33.3312970	-	0.9554529
	1.8623232		0.0558731	
listed_inDrama	-	27.2217306	-	0.2691130
	30.0979808		1.1056601	
listed_inDrama, Family	69.5949443	28.8668087	2.4108984	0.0160748
listed_inDrama, Family, Fantasy	52.3848005	26.3722591	1.9863600	0.0472387
listed_inDrama, Family, Historical	37.4750183	33.4117841	1.1216108	0.2622703
listed_inDrama, Family, Musical	85.9297922	33.3393283	2.5774302	0.0100818
listed_inDrama, Family, Mystery	39.2414098	33.3537594	1.1765213	0.2396386
listed_inDrama, Fantasy	67.3507948	28.7990999	2.3386424	0.0195305
listed_inDrama, Fantasy, Mystery	-	33.4932403	-	0.9254214
	3.1359097		0.0936281	
listed_inDrama, Historical	39.3055835	29.0924509	1.3510578	0.1769520
listed_inDrama, Historical, Sports	69.9825224	28.9058029	2.4210544	0.0156350
listed_inDrama, Music	85.1607218	28.8665770	2.9501496	0.0032428
listed_inDrama, Music, Romance	45.0390045	33.2093200	1.3562158	0.1753061
listed_inDrama, Mystery	80.7012443	33.2100885	2.4300220	0.0152555
listed_inDrama, Parody, Science Fiction	-	33.3599314	-	0.3552088
	30.8552378		0.9249191	
listed_inDrama, Romance	44.6034194	33.6008941	1.3274474	0.1846334
listed_inDrama, Sports	82.8711368	28.7951711	2.8779526	0.0040793
listed_inFamily, Fantasy	84.8389211	33.3345328	2.5450761	0.0110597
listed_inFamily, Fantasy, Musical	90.5971355	26.3768520	3.4347213	0.0006151
listed_inFamily, Fantasy, Mystery	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inFamily, Fantasy, Romance	43.7534561	33.3795229	1.3107873	0.1902005
listed_inFamily, Game Show / Competition	3.4376471	25.7415546	0.1335447	0.8937868
listed_inFamily, Game Show / Competition, Reality	0.0390045	33.2093200	0.0011745	0.9990631

term	estimate	std.error	statistic	p.value
listed_inFamily, Kids	18.5439185	33.3561818	0.5559365	0.5783662
listed_inFamily, Lifestyle, Reality	50.2675524	28.9011651	1.7392916	0.0822605
listed_inFamily, Music, Variety	32.6039423	28.7607125	1.1336278	0.2571953
listed_inFamily, Variety	36.5011609	28.9240722	1.2619648	0.2072263
listed_inFantasy	27.5171446	28.8509071	0.9537705	0.3404073
listed_inFantasy, Kids, Mystery	18.3360338	33.3612209	0.5496212	0.5826897
listed_inFantasy, Musical	50.9828263	27.1285989	1.8793019	0.0604649
listed_inFantasy, Musical, Thriller	8.7792534	33.2087222	0.2643659	0.7915470
listed_inFantasy, Variety	49.0390045	33.2093200	1.4766639	0.1400491
listed_inGame Show / Competition, Sports, Variety	-	33.2082952	-	0.9635124
listed_inHistorical, Musical	1.5195023		0.0457567	
listed_inKids	37.6571791	28.9011494	1.3029648	0.1928568
	-	33.2121378	-	0.9803487
	0.8182579		0.0246373	
listed_inKids, Music	18.1942992	29.2439738	0.6221555	0.5339673
listed_inLifestyle	-	33.4932403	-	0.9491637
	2.1359097		0.0637714	
listed_inMovies	7.5439335	27.1580059	0.2777794	0.7812334
listed_inMusic, Reality	-	33.4934778	-	0.9192649
	3.3956609		0.1013827	
listed_inMusical	38.7870546	33.3312744	1.1636835	0.2448021
listed_inMusical, Romance, Western	137.3451428	33.4144197	4.1103555	0.0000424
listed_inReality	-	26.4466776	-	0.9698711
	0.9991165		0.0377785	
listed_inScience Fiction	18.4779674	28.7953389	0.6416999	0.5212004
listed_inSeries	-	27.6928409	-	0.9450611
	1.9087521		0.0689258	
listed_inSports	87.2597511	33.2080390	2.6276695	0.0087157
listed_inTalk Show	-	33.3312744	-	0.9709774
	1.2129454		0.0363906	
listed_inVariety	22.5195023	33.2082952	0.6781288	0.4978312

```
glance(model_multi)
```

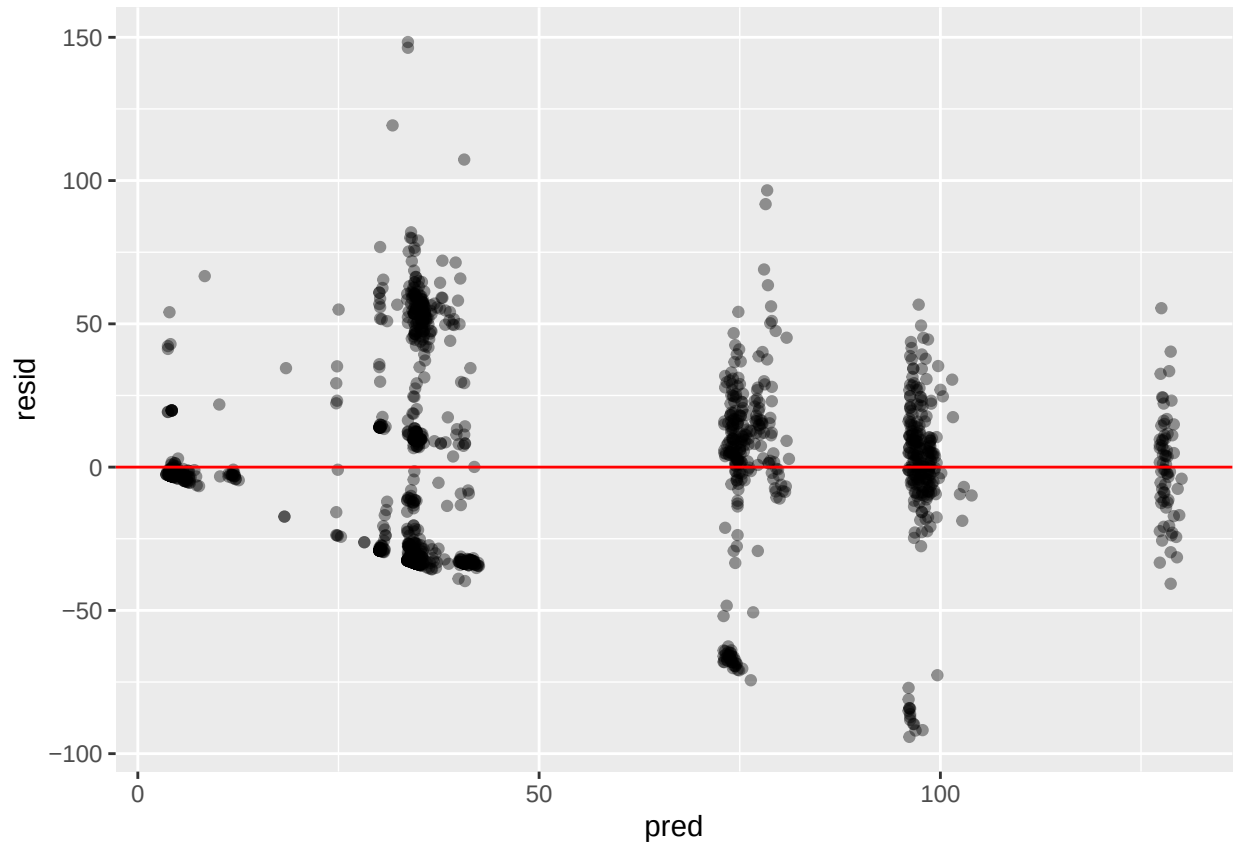
r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	cdf.residual	nobs
0.8049636	0.7456276	23.481571	13.56619	0	338	-	13561.8115356	78612587.7	1111	1450	6440.904

6.)

```
disney_pred <- disney %>%
  add_predictions(model_multi1, var = "pred") %>%
  add_residuais(model_multi1, var = "resid")
```

7.)

```
ggplot(disney_pred, aes(x = pred, y = resid)) +  
  geom_point(alpha = 0.4) +  
  geom_hline(yintercept = 0, color = "red")
```



```
ggplot(disney_pred, aes(x = pred, y = duration_num)) +  
  geom_point(alpha = 0.4) +  
  geom_abline(color = "blue")
```

